Comparative Analysis of Self-Supervised Learning Frameworks for Vision-Based PM2.5 Air Quality Estimation

CSE475 — Computer & Cyber Security | Section 02, Group 04

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Dataset — PM25Vision

Dataset Overview

Total Samples: 11,219 images
Task: 6-class AQI severity classification
Backbone (SSL): EfficientNet-B0 (0.394 GFLOPS)

Preprocessing

- RGB/HSV histograms for brightness & contrast
- Perceptual hashing to remove duplicates
- Laplacian variance for sharpness estimation
- Spatial crop/flip safe; color jitter constrained

Per-Class Distribution

AQI Class	Severity	Approx. Samples
0 – 50	Good	2,410
51 – 100	Moderate	2,410
101 – 150	Unhealthy for Sensitive Groups	2,410
151 – 200	Unhealthy	1,097
201 – 300	Very Unhealthy	482
301 – 600	Hazardous	2,410

Related Work

CNN

CNN-Based Vision Estimation

End-to-end CNNs infer AQI from pixel-level atmospheric degradation (contrast, color shift, scattering). Earlier approaches used dark channel priors & image entropy.

FSL

Few-Shot Learning (Prototypical Networks)

Metric-space approach classifies AQI by distance to class prototypes, retaining >92% accuracy with very few labeled examples.

LSTM

Hybrid CNN + LSTM Architectures

ResNet-LSTM and EfficientNet-BiLSTM combine spatial extraction with temporal modeling. HDR + weather metadata improves PM2.5 precision on mobile.

SSL

Self-Supervised Learning (SimCLR, MoCo → BYOL, Barlow Twins, DINO)

Contrastive methods (SimCLR/MoCo) require huge batches. Non-contrastive BYOL & Barlow Twins overcome this. DINO's multi-crop self-distillation captures local haze textures best.

RF

CNN + Random Forest Ensemble

CNN extracts edge & color features; Random Forest handles regression — reduces MAE by up to 11.11% by preventing overfitting.

Supervised Baselines — Best Split (90:10)

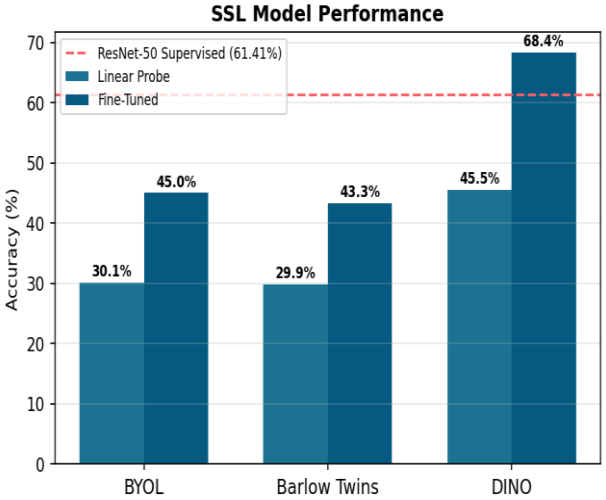
All models trained on 9 split ratios (10:90 → 90:10). Best results at 90:10 split shown below.

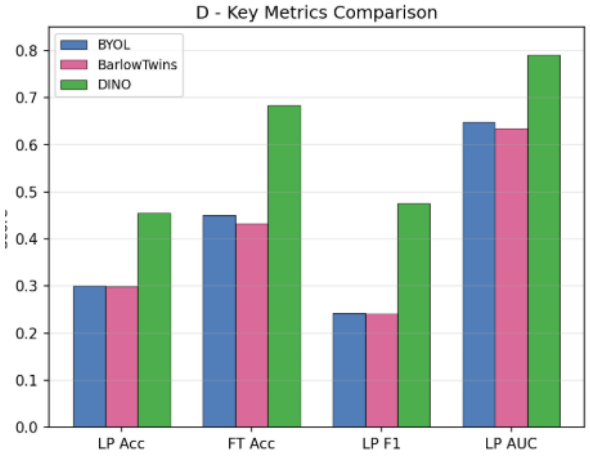
Model	Accuracy	Macro F1	Precision	Recall	Split
ResNet-50	61.44%	0.64	0.65	0.63	70:30
EfficientNet-B0	60.65%	0.64	0.65	0.63	80:20
GoogLeNet (Inception-V3)	59.54%	0.63	0.64	0.64	90:10
MobileNetV2	60.78%	0.62	0.66	0.63	80:20

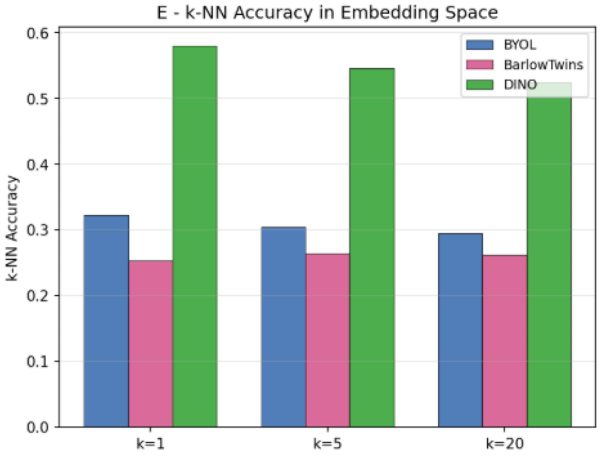
Self-Supervised Learning (SSL) — All Frameworks

All SSL models use EfficientNet-B0 backbone, pre-trained for 30 epochs without labels, then evaluated via Linear Probe & Full Fine-Tuning.

SSL Method	Linear Probe Acc.	Fine-Tuned Acc.	LP Macro F1	Paradigm
BYOL	30.12%	45.05%	0.24	Bootstrap (EMA)
Barlow Twins	29.86%	43.32%	0.24	Redundancy Reduction
DINO ★ Best	45.50%	68.36%	0.47	Self-Distillation (Multi-crop)



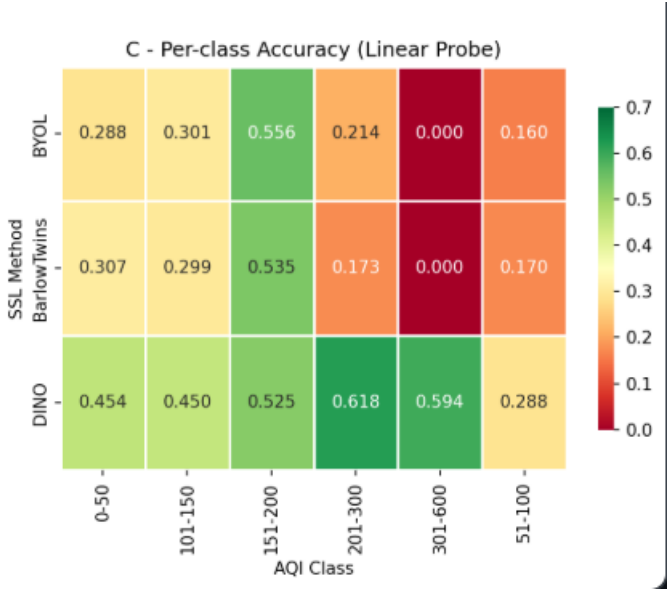




Performance Across Train:Test Splits

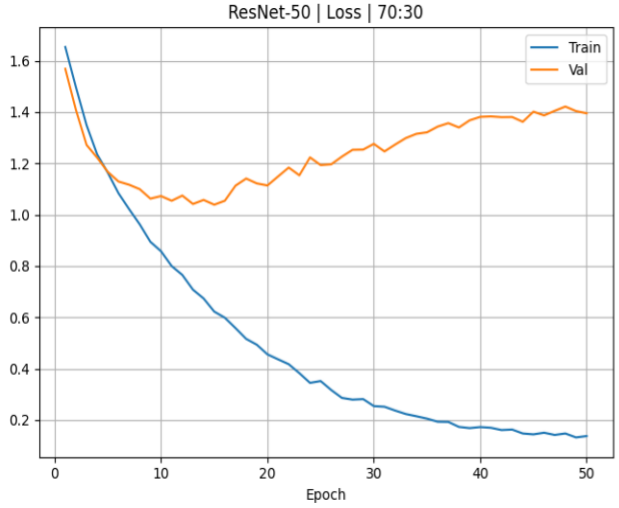
1	Split	BYOL	BarlowTwins	DINO
2	90:10	0.3091	0.3017	0.4749
3	80:20	0.3012	0.2986	0.455
4	70:30	0.295	0.29	0.4373
5	60:40	0.288	0.2926	0.4137
6	50:50	0.2851	0.28	0.3946
7	40:60	0.2706	0.275	0.3689
8	30:70	0.2686	0.2691	0.3389
9	20:80	0.25	0.2539	0.2989
10	10:90	0.2305	0.2354	0.2437

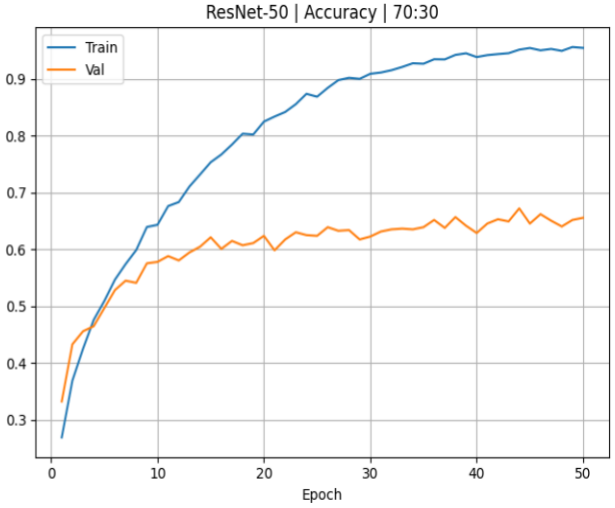
1	SSL	Backbone	LP_Accuracy	LP_F1	LP_AUC	LP_Precision	LP_Recall	FT_Accuracy	FT_F1
2	BYOL	efficientnet_b0	0.3012	0.2424	0.6479	0.2473	0.2531	0.4505	0.4607
3	BarlowTwins	efficientnet_b0	0.2986	0.2409	0.6348	0.2653	0.2473	0.4332	0.45
4	DINO	efficientnet_b0	0.455	0.4753	0.7902	0.4652	0.4883	0.6836	0.7204

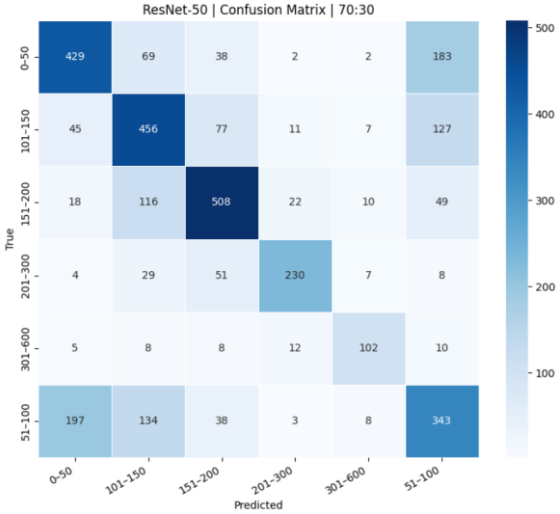


Supervised Best Model — ResNet-50 Split Analysis

61.41%	0.6431	7,067	0.6549
Best Accuracy (70:30 split)	Macro F1 (70:30 split)	Training Images (70:30)	Macro precesion 70:30

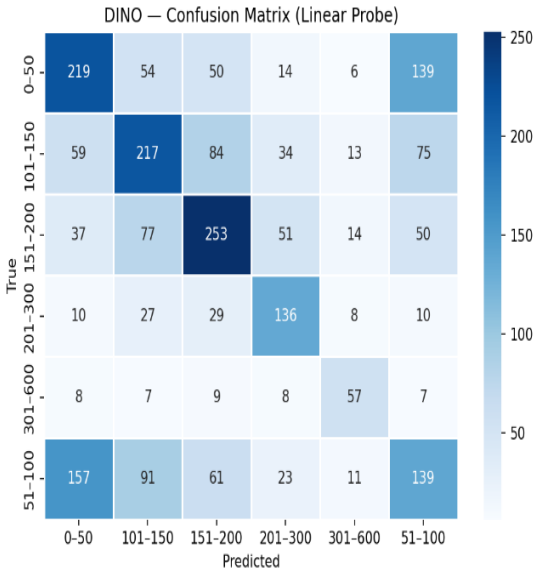


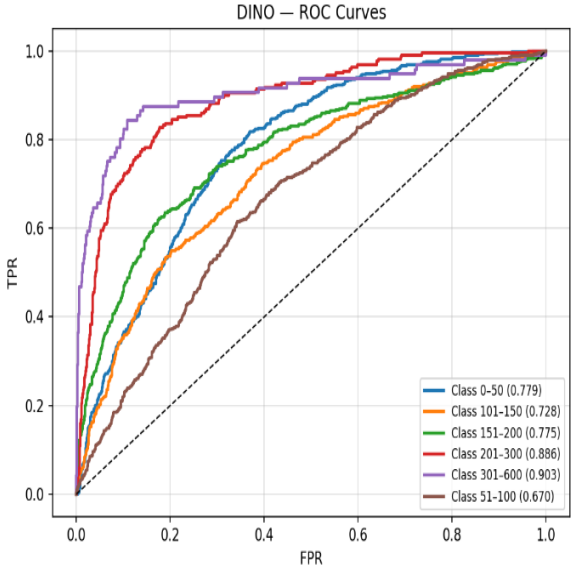


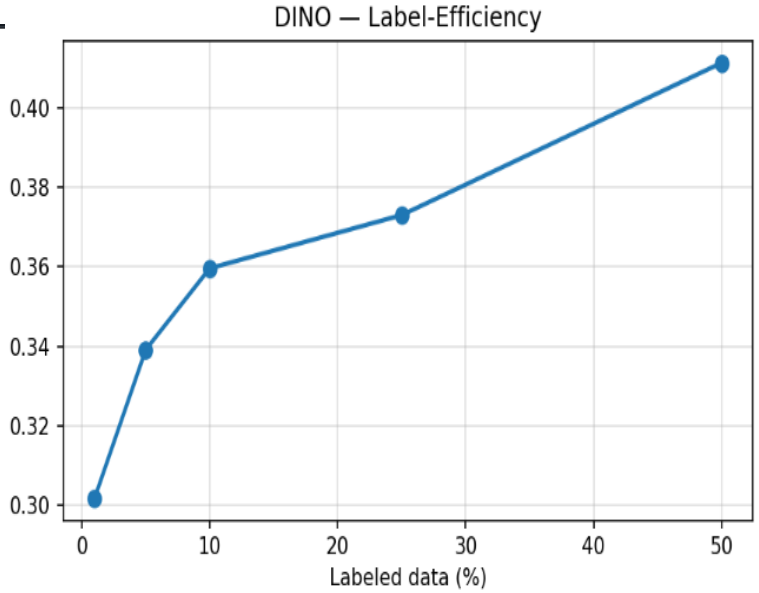


DINO — SSL Performance Charts

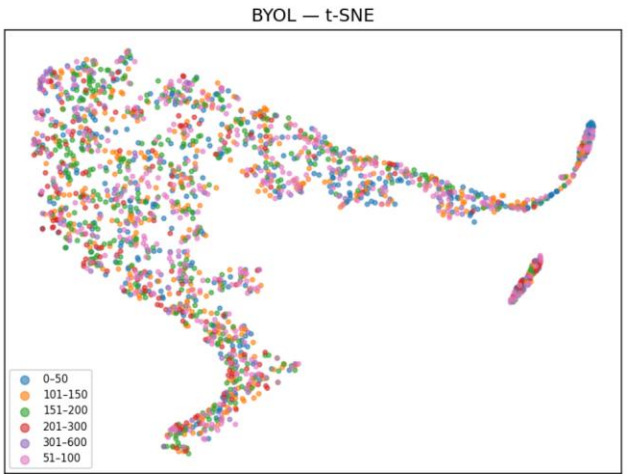
45.50%	68.36%	59.38%	0.4753
Linear Probe Accuracy	Fine-Tuned Accuracy	Hazardous Class Accuracy (301-600)	Macro f1

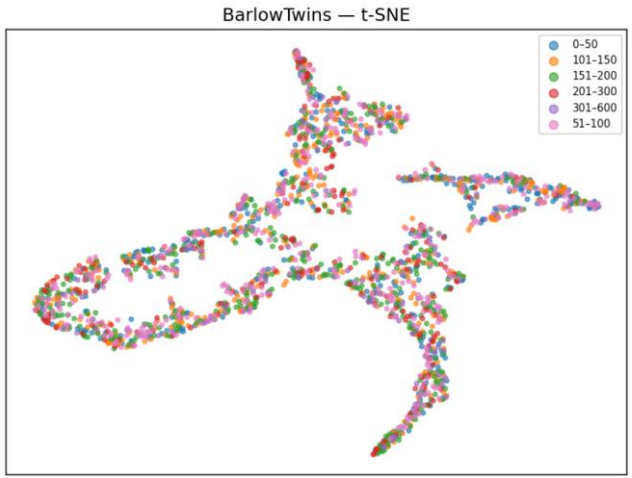


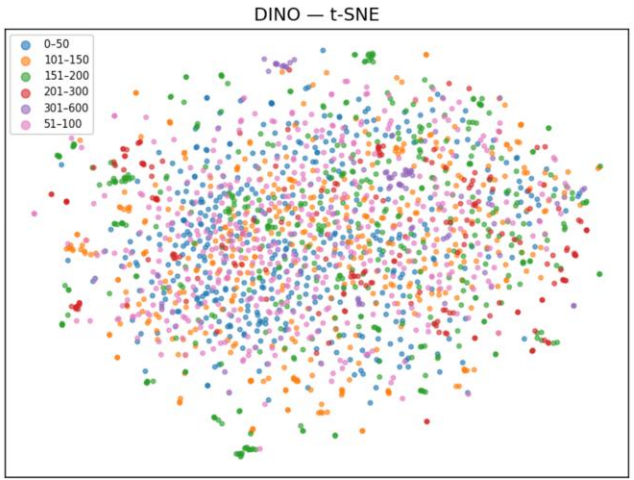




Embedding Space Analysis — t-SNE for All SSL Methods







SSL Model	Silhouette Score	Interpretation	k-NN Acc. (k=1)
BYOL	−0.1544	Highly overlapping, no AQI structure	32.26%
Barlow Twins	−0.0459	Slightly better, still continuous overlap	25.31%
DINO ★ Best	−0.0305	Best structural separation among all SSL	57.98%

Note: All models show negative silhouette scores — expected because AQI haze is a continuous visual spectrum, not discrete object categories.